Faisal Mujawar

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Education

Royal Holloway, University of London Expected: June 2026

B.Sc. in Computer Science

- Grade for First-year: First Class.

Projects

Coreboot/Libreboot: Thinkpad X201 September 2024

- Utilised **CH341A programmer + SOIC8 clip** to flash the **EEPROM** with custom settings and replaced the bios from stock to Coreboot for performance and security benefits, had to manually configure the **Coreboot+SeaBIOS** to be optimised for the tablet version of this thinkpad.
- Diagnosed and recovered **bricked firmware** by analyzing system registers and comparing them to previous images.

RISC-V Emulator: RISVEmu October 2024

- Developed a **RISC-V emulator** in **C++**, emulating the functionality of the **RV32I** instruction set.
- Added support for emulated syscalls to enable user I/O, allowing a simple text-based "choose-your-own-adventure" game. assembled in **RV32I**, to run on the emulator.
- The reason behind is to gain a deeper understanding of modern CPU architecture and RISC ISAs.

University Management System: ImaginaryUni December 2024

- Designed and implemented a university management system using **Spring Boot** for the backend and **react+tailwind** for the frontend.
- Integrated features like student data management, course enrollment, and staff administration, and observers to update student/staff when changes corresponding to them occur.
- UML-based designs to ensure a maintainable and efficient codebase.

Hackaway V8 MLH Hackthon: CVision - Chrome Extension January 2025

- Developed a resume analysis Chrome extension using **NLP** and machine learning to identify skill gaps with **82% accuracy**, implementing text processing and **scikit-learn** for job description matching
- Engineered a **Flask API** for resume parsing and keyword extraction, integrated with **Next.js** frontend for interactive visualizations of skill matches/gaps
- Implemented Chrome extension core functionality using **Manifest V3**, content scripts for page scraping (most) sites.
- Designed PDF parsing pipeline supporting multiple formats (PDF/DOCX) using **pdfminer** and **python-docx**, achieving 90% successful parse rate across test cases

[ONGOING] NES Emulator: NEmuS January 2025

- Developing a Nintendo Entertainment System (Ricoh 2A03) emulator in **C++**, accurately emulating the NES hardware architecture.
- The idea behind is to gain fundamental knowledge of system architecture emulation, especially **MMIO, system memory maps, interrupt handling, etc.** which will be helpful for emulating more complex architectures like that of the PS1.
- Ongoing extensions: **Picture Processing Unit (PPU) and Audio Processing Unit (APU), and Bus interconnect**.

Skills & Interests

Technical Skills: C/C++, Python(BeautifulSoup4, Pandas, Matplotlib, scikit-learn), MySQL/Postgres, Java(Springboot, JUnit, Mockito)

Extracurricular Activities:

- Hackathons: Royal Hackaway v7, Codedex 2024, Headstarter Hackathon, etc.
- Lead volunteer at Saint John's Church community fridge

Experience

Maths Mentor, CeDAS October 2024 - Present

Royal Holloway, University of London

- Assisted 1st-year Computer Science students with **mathematics and machine fundamentals** during weekly drop-in sessions, improving their understanding of key concepts.
- Developed tutoring skills by breaking down complex mathematical problems into manageable steps.
- Provided support to students from diverse academic backgrounds, helping them boost confidence and performance.

Research Assistant - EPMS February 2025 - Present

Royal Holloway, University of London

- Researching AI implementation in anti-cheat systems, focusing on machine learning techniques to detect and prevent improper conduct in gaming and academic environments
- Working under Professor Konstantinos Markantonaki contributing to cutting-edge security research initiatives